

Unbounded spreading in Harmonic Grammar with gradient representations¹

This paper presents an initial analysis of unbounded spreading in Harmonic Grammar as the redistribution of gradient activity. We demonstrate that the model accounts for local (=myopic) blocking effects, is consistent with parallel and serial frameworks, and can be expanded to attested model non-myopic effects in Tutrugbu.

Keywords: unbounded spreading, Harmonic Grammar, gradient symbolic representations

“Why, I feel all thin, sort of stretched, if you know what I mean: like butter that has been scraped over too much bread.”

Bilbo Baggins, *The Fellowship of the Ring*

1. INTRODUCTION

Unbounded feature spreading raises theoretical questions in constraint-based formalisms like Optimality Theory (OT; Prince & Smolensky 1993/2004; McCarthy & Prince 1993b) and Harmonic Grammar (HG; Legendre et al. 1990a; b; Pater 2009; 2016). In particular, whether a trigger adjacent to a string of targets constitutes a single locus of violation itself or multiple loci counting the targets. As a concrete example, nasal segments in Tuparí (tpr, Tupian) trigger progressive nasalization (1), with some morphological restrictions (Singerman 2018: 395-396). Examples such as (1g) illustrate that spreading is strictly progressive *[tẽ.ẽẽ.mã.toa], with the exception of the pronominal prefixes /o-/ ‘1sg’, /e-/ ‘2sg’, /i-/ ‘3’, which are exceptionally subject to regressive nasalization (Singerman 2016: 463). As in these examples, morpheme-final /t/ and /n/ flap before vowel-initial morphemes (Singerman 2016: 459). In OT and HG, unfaithful

[1] removed for review

mappings such as the verb /aot/ ‘leave’ surfacing with nasal vowels (1b-d) must improve on phonotactic constraints.

(1) Nasalization in Tuparí (Singerman 2016: 454-455, 463-464, 467)

(a)	/o-aot-a/	[wao.ra]	‘I leave’
(b)	/o-m-aot-a/	[õ.mãõ.řã]	‘make me leave’
(c)	/o-m-aot-a-p/	[õ.mãõ.řãm]	‘making me leave’
(d)	/o-m-aot-a-t/	[õ.mãõ.řãn]	‘made me leave’
(e)	/ko-roro/	[ko.ro.ro]	‘eat, drink-NMLZ’
(f)	/nẽ-roro/	[nẽ.řõ.řõ]	‘do, make-NMLZ’
(g)	/te-we-m-ato-a/	[te.we.mã.toa]	‘bathe one another’

Under a single locus hypothesis, a faithful form like *[õ.mao.ra] might violate a constraint against a nasal segment followed by an oral segment, as would unfaithful forms with partial spreading such as *[õ.mãõ.ra]. A multiple locus hypothesis might instead penalize all oral segments following a nasal segment, thus assigning four violations to the faithful form *[õ.mao.ra], fewer to those with partial spreading, and none to the surface form [õ.mãõ.řã].

Single and multiple locus approaches both adequately model mappings that result in total nasalization such as /o-m-aot-a/ → [õ.mãõ.řã]. However, they are distinguished by their treatment of partial spreading. In Tuparí, voiceless obstruent onsets block progressive nasalization, as in [mã.ka] *[mã.kã] (2d); examples follow on the next page. In coda position, obstruents nasalize following nasal vowels as in [mãŋ] *[mãk] ‘hand it over!’ (Singerman 2016: 470).

Under the single locus hypothesis, spreading up to a blocker raises a theoretical challenge. If faithful forms like [ma.toa] and unfaithful forms like [mã.toa] fare equally with respect to the phonotactic constraint, then the latter is harmonically bounded because it is needlessly unfaithful (Wilson 2003; 2006; McCarthy 2010); the tableau in 3 illustrates. The faithful candidate violates AGREE(nasal) (Lombardi 1999; Baković 2000) as does the intended winner, both having a nasal segment adjacent to a non-nasal segment.

- (2) Voiceless obstruent onsets block nasalization in Tuparí (Singerman 2016: 462, 465, 467-468, 470)

- (a) /ãpe/ [ã.pe] ‘comb’
 (b) /m-ato-a/ [mã.toa] ‘bathe someone else’
 (c) /e-m-aoros-a-p/ [ẽ.mãõ.řõ.sap̚] ‘making you arrive’
 (d) /māk-a/ [mã.ka] ‘send, hand over’
 (e) /e-õ-ʔet-a-t/ [ẽõ.ʔe.rat̚] ‘made you sleep’

Total nasalization satisfies AGREE(nasal), but is ruled out by onset faithfulness (Beckman 1998). Because the intended winner does not improve on the phonotactic constraint, its faithfulness violations prevent it from winning.

- (3) Non-local blocking in OT

		IDENTONSET(son)	AGREE(nasal)	IDENT(nasal)
/m-ato-a/				
☐ a. [ma.toa]			1	
☹ b. [mã.toa]			1	W 1
c. [mã.řõã]	W 1	L		W 4

Under the multiple locus hypothesis, blocking effects are difficult to derive without enriching the constraints. For example, to avoid predicting *[mã.kã], it may be necessary to define loci as possible targets up to a blocker, which is impossible with alignment constraints (McCarthy & Prince 1993a; McCarthy 2003) which have been used to motivate spreading (Piggott 2000; Jurgec 2011). Further, as Leduc & McCollum (2023) highlight, constraints like AGREE are necessary to motivate both feature spreading and deletion, which cannot be modeled by other multiple locus constraints like SHARE (McCarthy 2010; Mullin 2011).

The present paper presents an analysis assuming the single locus hypothesis

in HG. In OT, the problem of partial spreading has been solved by differentiating between loci by either rewarding derived loci (Wilson 2000; 2003) or by penalizing them in proportion to their position in the string (Eisner 2000; 2002; Finley 2008; Lamont 2022). In HG, Kimper (2011) implemented a scaling factor to reward every instance of spreading. Our proposal similarly graduates loci such that a faithful form like *[ma.ka] is dispreferred to an unfaithful form like [mã.ka]. Based on recent developments in the research on gradient representations, we argue for a representational solution. We demonstrate that this approach derives unbounded spreading, models local blocking, is consistent with parallel and serial models, and is sufficiently expressive to be typologically adequate.

2. SPREADING AS REDISTRIBUTING ACTIVITY

Before presenting the technical details of the analysis, allow us to abuse the metaphor of feature spreading to provide an intuition. Spreading a pat of butter on a waffle redistributes the butter across its pockets, minimizing the difference in butter between adjacent pockets. The further the pat is spread, the more the difference between the butter's frontier and adjacent dry pockets decreases. Our analysis works analogously, motivating spreading by penalizing adjacent segments that disagree in some feature.

We follow Tebay (2023) in generalizing gradient symbolic representations (Smolensky et al. 2014; Smolensky & Goldrick 2016; Rosen 2016; Cho et al. 2017; Zimmermann 2018a; b; 2019; 2021; Kushnir 2019; Walker 2020) to autosegments (Goldsmith 1976; Leben 1973; 2011). Gradient representations imbue phonological objects with a measure of activity specified as a rational number between zero and one, inclusive. Following Zimmermann, we treat activity as an abstraction without any phonetic interpretation: segments either have 0 activity of a feature or > 0 activity. Its analytic value is its providing a means of distinguishing strong and weak segments by according them more or less severe constraint violations.

As a concrete example, some word-final vowels in Tuparí idiosyncratically delete before the theme vowel /-a/. The /i/ in /si/ deletes before the theme vowel (4a), but not the /i/ in /ni/ or /mi/ (4b-c).

(4) Lexically conditioned vowel deletion in Tuparí (Singerman 2018: 387)

- (a) [si] [sa] ‘spear, kill, sting’
 (b) [nĩ] [nĩ.ã] ‘be embarrassed’
 (c) [mĩ] [mĩ.ã] ‘stab, pierce’

Vowels in words like /si/ are weaker and more susceptible to deletion. This translates to their being underlyingly specified with less activity than other vowels. In the tableaux below, the vowel in /si/ only has activity 0.5, whereas the vowel in /ni/ has activity 1. In order to save space, we only write activities below 1.

(5) Exceptional deletion as low activity

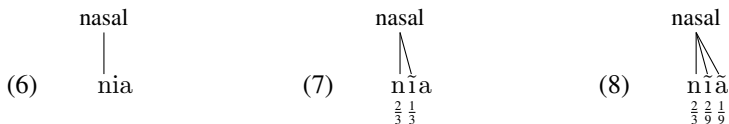
		*VV	Max	$\mathcal{H}$
		4	5	
/si-a/ .5	a. [si.a] .5	-.75		-3
	☞ b. [sa]		-.5	-2.5
	c. [si] .5		-1	-5
/ni-a/ ☞	a. [nĩ.ã]	-1		-4
	b. [nã]		-1	-5
	c. [nĩ]		-1	-5

The faithful forms violate a phonotactic constraint against adjacent vowels *VV. Following Zimmermann (2019), constraint violations are proportional to the average activity of their loci. Accordingly, *VV assigns faithful [si.a]_{.5} a violation of 0.75 (average of 0.5 and 1) and faithful [nĩ.ã] a violation of 1 (average of 1 and 1). Deleting the weak /i/_{.5} only incurs a violation of 0.5 of Max, but deleting any other vowel incurs a full violation of 1. As the tableaux illustrate, deleting the weak /i/ is preferred to tolerating it in hiatus, but deleting full vowels is too

expensive.

As in this example, gradient representations allow for gradient constraint violations: the weak /i/ incurs a smaller violation of *VV than a full vowel. Deleting it as in [sa] removes the violation, but removing activity would also improve on *VV albeit without satisfying it. This is our proposed mechanism behind autosegmental spreading. The edge of an autosegmental span incurs a violation of AGREE in proportion to its activity. Spreading distributes activity across a larger number of segments, thus reducing the relative violation.

We model spreading as shifting some activity from one segment onto an adjacent string of segments. Specifically, we allow GEN to fission off a fixed proportion of activity, which we arbitrarily set to $\frac{1}{3}$. Unbounded spreading is modeled by recursively fissioning activity. Thus, mapping /nia/ onto the faithful form [ni.a] keeps all of the nasal activity on the head [n] (6), spreading onto the adjacent vowel fissions one third of the activity (7), and spreading further fissions one third of the fissioned third (8). Candidates that alter which segment is the head of the span can be ruled out by faithfulness constraints following Key (2007), Key & Bickmore (2014), and Tebay (2022), and are not considered below.



Spreading improves on an AGREE constraint, which is interpreted gradiently. The faithful form in (6) incurs $\frac{1}{2}$ a violation (average of 1 and 0), the partial spreading form in (7) incurs $\frac{1}{6}$ a violation (average of $\frac{1}{3}$ and 0), and the full spreading form in (8) satisfies AGREE(nasal).

(9) AGREE(F)

For every pair of adjacent segments x, y where x is linked to a feature F and y is not, assign $\frac{a_x+0}{2}$ violations where a_x is the activity of F associated to x .

Spreading violates faithfulness constraints against adding and removing activity from a segment (cf. constraints against adding and removing autosegmental

association lines à la Yip 2002). The forms (7-8) above remove $\frac{1}{3}$ of the activity of the nasal feature from /n/ and shift it onto the vowel chain. Thus, they both incur $\frac{1}{3}$ violations of both $\text{MAX}(\text{activity})/\text{nasal}$, which penalizes removing activity from a segment, and $\text{DEP}(\text{activity})/\text{nasal}$, which penalizes adding activity to a segment. In general, spreading incurs equal violations of these constraints as it neither removes nor creates activity.

(10) $\text{DEP}(\text{activity})/\text{F}$

For every segment with a_i activity of the feature F in the input and a_o activity in the output, assign $a_o - a_i$ violations if $a_i < a_o$.

(11) $\text{MAX}(\text{activity})/\text{F}$

For every segment with a_i activity of the feature F in the input and a_o activity in the output, assign $a_i - a_o$ violations if $a_i > a_o$.

To control direction of spreading, we introduce two phonotactic constraints: *FALLINGACTIVITY penalizes falls in activity, preferring strictly regressive spreading and *RISINGACTIVITY penalizes rises in activity, preferring strictly progressive spreading.

(12) *FALLINGACTIVITY

For every autosegmental span with activity a , assign a violations if there are adjacent segments $x \triangleleft y$, where x has more activity than y .

(13) *RISINGACTIVITY

For every autosegmental span with activity a , assign a violations if there are adjacent segments $x \triangleleft y$, where x has less activity than y .

$\text{DEP}(\text{activity})/\text{F}$, $\text{MAX}(\text{activity})/\text{F}$, *FALLINGACTIVITY , and *RISINGACTIVITY are all antagonistic to spreading. Crucially, the violations of all four constraints are bounded. Violations of the two faithfulness constraints are bounded at $\frac{1}{3}$ under unidirectional spreading and at $\frac{5}{9}$ under bidirectional spreading. The phonotactic constraints are bounded at 1 because this is the maximum activity of any feature. It is crucial to establish these bounds because without them, HG cannot model

unbounded spreading under the single locus hypothesis (O’Hara 2016). As an illustration, consider the tableau in 14. It is impossible to choose non-zero weights x and y that prefer the unfaithful candidate (14b) to the faithful candidate (14a) for all inputs. The faithful candidate’s harmony is constant, while the harmony of complete nasalization grows in the length of the input. Eventually, the cumulative violations of IDENT(nasal) will exceed the violation of AGREE(nasal), predicting spreading only in strings up to a fixed length.

(14) Bounded spreading with categorical representations

/ma...a/	AGREE(nasal) x	IDENT(nasal) y	$\mathcal{H}$
a. [ma...a]	-1		$-x$
b. [mã...ã]		$-n$	$-ny$

By bounding the violations of the constraints antagonistic to spreading, our approach avoids this pathology. The tableau in (15) illustrates the mapping /ni/ $\rightarrow$ [nĩ], which satisfies AGREE(nasal) by spreading onto the adjacent vowel.

(15) /ni/ $\rightarrow$ [nĩ] ‘be embarrassed’ (Tupari)

nasal ni	AGREE(nasal) 10	*FALLINGACTIVITY 1	DEP(link)/nasal 3	MAX(link)/nasal 3	$\mathcal{H}$
a. nasal [ni]	$-\frac{1}{2}$				-5
b. nasal ↙ [nĩ] ↘ $\frac{2}{3} \quad \frac{1}{3}$		-1	$-\frac{1}{3}$	$-\frac{1}{3}$	-3

The faithful candidate incurs $\frac{1}{2}$ violations of AGREE(nasal) and is dispreferred to the unfaithful candidate which incurs 1 violation of *FALLINGACTIVITY and $\frac{1}{3}$ violations of the faithfulness constraints.

Because spreading onto more segments does not change the violations of *FALLINGACTIVITY, DEP(link), or MAX(link), the same constraint weights derive unbounded spreading. The tableau in (16) illustrates the mapping /ne-oro/ → [nẽ.řõ.řõ], which nasalizes five segments.

(16) /ne-oro/ → [nẽ.řõ.řõ] ‘do, make-NMLZ’ (Tuparí)

nasal ne-oro	AGREE(nasal) 10	*FALLINGACTIVITY 1	DEP(link)/nasal 3	MAX(link)/nasal 3	$\mathcal{H}$
a. nasal [ne.ro.ro]	$-\frac{1}{2}$				-5
b. nasal ↙ [nẽ.ro.ro] $\frac{2}{3} \quad \frac{1}{3}$	$-\frac{1}{6}$	-1	$-\frac{1}{3}$	$-\frac{1}{3}$	$-4\frac{2}{3}$
c. nasal ↙ ↘ [nẽ.řõ.ro] $\frac{2}{3} \quad \frac{2}{9} \quad \frac{1}{9}$	$-\frac{1}{18}$	-1	$-\frac{1}{3}$	$-\frac{1}{3}$	$-3\frac{5}{9}$
d. nasal ↙ ↘ ↘ [nẽ.ř õ.řõ] $\frac{2}{3} \quad \frac{2}{9} \quad \frac{2}{27} \quad \frac{1}{27}$	$-\frac{1}{54}$	-1	$-\frac{1}{3}$	$-\frac{1}{3}$	$-3\frac{5}{27}$
e. nasal ↙ ↘ ↘ ↘ [nẽ.ř õ.ř õ] $\frac{2}{3} \quad \frac{2}{9} \quad \frac{2}{27} \quad \frac{2}{81} \quad \frac{1}{81}$	$-\frac{1}{162}$	-1	$-\frac{1}{3}$	$-\frac{1}{3}$	$-3\frac{5}{81}$
f.  nasal ↙ ↘ ↘ ↘ ↘ [nẽ.ř õ.ř õ] $\frac{2}{3} \quad \frac{2}{9} \quad \frac{2}{27} \quad \frac{2}{81} \quad \frac{2}{243} \quad \frac{1}{243}$		-1	$-\frac{1}{3}$	$-\frac{1}{3}$	-3

In general, if it optimal to spread onto one segment to satisfy AGREE, then it is optimal to spread onto arbitrarily many segments. AGREE is defined locally over adjacent segments, and the harmonic improvement gained by satisfying it does not depend on the length of the string. Likewise, because the cost of the antagonistic constraints is constant, their penalty does not depend on the length of the string. By default, spreading is therefore unbounded, unless other constraints intervene (McCollum 2022; McCollum & Kavitskaya 2018; 2022).

The other important property that the tableau above illustrates is that violations of AGREE fall exponentially with the length of the autosegmental span. As every segment has a third of the activity of its precedent, if it is adjacent to a non-nasal segment, its violation is a third of what its precedent's would be.

(17) /m-ato-a/ → [mã.toa] ‘bathe someone else’ (Tuparí)

	nasal m-ato-a	IDENTONSET(son)	AGREE(nasal)	*FALLINGACTIVITY	DEP(link)/nasal	MAX(link)/nasal	$\mathcal{H}$
		3	10	1	3	3	
	nasal a. ma.toa		$-\frac{1}{2}$				-5
☞	nasal b. mã.toa $\frac{2}{3} \quad \frac{1}{3}$		$-\frac{1}{6}$	-1	$-\frac{1}{3}$	$-\frac{1}{3}$	$-4\frac{2}{3}$
	nasal c. mã. r̥ o̥ ã $\frac{2}{3} \quad \frac{2}{9} \quad \frac{2}{27} \quad \frac{1}{81}$	-1		-1	$-\frac{1}{3}$	$-\frac{1}{3}$	-6

This implies that spreading is harmonically improving even when it does not satisfy AGREE, allowing local blocking as in the tableau in (17; cf. the non-local effects in OT in tableau 3). Complete nasalization is blocked because it would change an obstruent in a syllable onset, violating the positional faithfulness

constraint. However, because spreading reduces the violation of $\text{AGREE}(\text{nasal})$, it is optimal to spread up to the blocker.

Further, while we have been assuming parallel evaluation, the gradual improvement with respect to $\text{AGREE}(\text{nasal})$ models spreading in serial HG (Pater 2012; Kaplan 2018). This is illustrated abstractly in the tableau in (18), where a nasal autosegment has already spread onto n segments. Provided that the weight of $\text{AGREE}(\text{nasal})$ is greater than twice the weight of $\text{DEP}(\text{link})/\text{nasal}$ and $\text{MAX}(\text{link})/\text{nasal}$, every additional instance of spreading is optimal even if it does not satisfy $\text{AGREE}(\text{nasal})$.

(18) $\text{m}\tilde{\text{a}}\dots\tilde{\text{a}}\text{a}\dots \rightarrow \text{m}\tilde{\text{a}}\dots\tilde{\text{a}}\tilde{\text{a}}\dots$, serial HG

nasal $\swarrow$ $\text{m}\dots\tilde{\text{a}}\text{a}\dots$ $\frac{2}{3} \quad \frac{1}{3^n}$	$\text{AGREE}(\text{nasal})$ 10	$*\text{FALLINGACTIVITY}$ 1	$\text{DEP}(\text{link})/\text{nasal}$ 3	$\text{MAX}(\text{link})/\text{nasal}$ 3	$\mathcal{H}$
a. nasal $\swarrow$ $\text{a. m}\dots\tilde{\text{a}}\text{a}\dots$ $\frac{2}{3} \quad \frac{1}{3^n}$	$-\frac{1}{2 \times 3^n}$	-1			$-1 \frac{5}{3^n}$
b. nasal $\swarrow$ $\text{a. m}\dots\tilde{\text{a}}\tilde{\text{a}}\dots$ $\frac{2}{3} \quad \frac{2}{3^{n+1}} \quad \frac{1}{3^{n+1}}$	$-\frac{1}{2 \times 3^{n+1}}$	-1	$-\frac{1}{3^{n+1}}$	$-\frac{1}{3^{n+1}}$	$-1 \frac{11}{3^{n+1}}$

In general, we have shown that feature spreading can be modeled as redistributing the activity of an autosegment. Spreading across an entire domain satisfies an AGREE constraint, and partial spreading reduces the violation by reducing the activity associated to segments at the edge of a span. Our implementation models unbounded spreading while avoiding problems with AGREE constraints in OT, HG, and serial models. In particular, because partial spreading is harmonically improving, serial derivations are possible and blocking effects are strictly local. As this work is preliminary, we have only shown one analysis as a proof of concept

that avoids problems like non-local blocking. Whether these emerge under other weightings of the constraints is an open question.

3. NON-LOCAL BLOCKING IN TUTRUGBU

The analysis above models feature spreading that depends only on the segment adjacent to the span, which Wilson (2003; 2006) dubs *myopic*. Myopic spreading is characterized by the local effect of blockers. As in Tuparí, nasalization spreads up to onset obstruents [ẽ.mãõ.řõ.sap̃] ‘making you arrive’ (2c), instead of being inhibited at a distance *[ẽ.mao.ro.sap̃]; see Walker (2011) for other examples. The analysis can be extended to model non-local effects (Jardine 2016; McCollum et al. 2020; Meinhardt et al. 2024), provided that feature spreading from the opposite direction renders blockers local. Computationally, phonological mappings that require information unboundedly distant in two directions can be modeled by two separate mappings operating in serial, one left-to-right and the other right-to-left (Heinz & Lai 2013; Meinhardt et al. 2024; Yolyan 2025). This section demonstrates the feasibility of such an approach; its empirical adequacy typologically is an open question.

Non-local blocking effects are attested in Tutrugbu (nyb, Niger-Congo); see McCollum & Essegbey (2018) and McCollum et al. (2020) for discussion. As in (19a), prefix vowels are [−ATR] by default, surfacing as such before roots without [+ATR] vowels. As in these examples, /ʊ/ and /ɪ/ surface as [ɔ] and [ɛ] in [−ATR] contexts and /a/ surfaces as [e] in [+ATR] contexts (McCollum et al. 2020: 218-219). Prefixes surface with [+ATR] vowels before roots with [+ATR] vowels (19b-c), having undergone regressive feature spreading. When the word-initial vowel is underlyingly high, [ATR] spreading is blocked by low vowels *[itɪbɛfɛ] (2d-g). This blocking effect is non-local because it depends on whether the initial vowel is underlyingly high and opaque because it depends on the underlying height of the initial vowel (compare 19c and 19g).

(19) ATR harmony in Tutrugbu (McCollum et al. 2020: 217, 220-221)

- | | | | |
|-----|---------------|------------|---------------------|
| (a) | /a-tí-ba-bá/ | [atɛ́babá] | ‘3SG-NEG-FUT-come’ |
| (b) | /a-tí-ba-ǰē/ | [etíbeǰē] | ‘3SG-NEG-FUT-grow’ |
| (c) | /ɔ-tí-ba-ǰē/ | [otíboǰē] | ‘2SG-NEG-FUT-grow’ |
| (d) | /ɪ-tí-ba-bá/ | [ɛtɛ́babá] | ‘1SG-NEG-FUT-come’ |
| (e) | /ɪ-tí-ba-ǰē/ | [ɛtɛ́baǰē] | ‘1SG-NEG-FUT-grow’ |
| (f) | /ɪ-ba-dɪ-wu/ | [ɛbadiwu] | ‘1SG-FUT-ITV-climb’ |
| (g) | /bu-ba-dɪ-wu/ | [bɔbadiwu] | ‘1PL-FUT-ITV-climb’ |

Given the opaque status of initial high vowels in Tutrugbu, it is necessary to adopt a stratal analysis; see Kiparsky (2024) for discussion of analyses along these lines and Tebay et al. (2025) for discussion of gradient representations across strata. We propose that [high] spreads progressively from initial underlying high vowels, raising underlying /a/ to [ə]. As McCollum & Essegbey (2018) point out, treating initial vowels in the language as exceptional triggers of spreading is independently supported by the language’s labial harmony, as in (19c) *[otíbeǰē] (McCollum & Essegbey 2020). A constraint against spreading [+ATR] onto schwa renders these vowels opaque in the following stratum. Lastly, [high] features delete unless linked to a [+ATR] vowel.

The tableaux below illustrate the core mechanisms of the analysis with the mapping /kɪ-ba-ǰē/ → [kɪbaǰē] ‘CL5-FUT-grow’ (McCollum et al. 2020: 220). In the first stratum, [high] spreads from the initial vowel onto all following prefixes, raising /a/ to ə; root faithfulness (McCarthy & Prince 1995) prevents the feature from spreading into the root /ǰē/. In order to restrict triggers to initial syllables, we rely on a positional variant of AGREE that only assigns violations when the loci involve the first syllable of the word. In this stratum, AGREE(ATR) must have very little weight to prevent [+ATR] spreading leftwards from the root.

(20) Stratum 1

high ATR ki-ba-fē	AGREE _{INITIAL} (high)	*FALLING _{ACTIVITY}	DEP(link)/high	MAX(link)/high	
	10	1	3	3	$\mathcal{H}$
high ATR a. ki-ba-fē	$-\frac{1}{2}$				-5
high ATR b. ki-bə-fē $\frac{2}{3}$ $\frac{1}{3}$	$-\frac{1}{3}$	-1	$-\frac{1}{3}$	$-\frac{1}{3}$	$-4\frac{2}{3}$

In the second stratum, [+ATR] spreads regressively from the root, but is blocked by ə. We model this with a faithfulness constraint DEP(link)/ə that penalizes linking features to schwa. Thus, [+ATR] may only spread onto strings of high vowels adjacent to the root when the word-initial vowel is high. In all other cases, the entire string of prefixes will be [+ATR].

(21) Stratum 2

high ATR ki-bə-fē $\frac{2}{3}$ $\frac{1}{3}$	DEP(link)/ə	AGREE(ATR)	*RISING _{ACTIVITY}	DEP(link)/ATR	MAX(link)/ATR	
	27	10	1	3	3	$\mathcal{H}$
high ATR a. ki-bə-fē		$-\frac{1}{2}$				-10
high ATR b. ki-bə-fē	$-\frac{2}{9}$		-1	$-\frac{1}{3}$	$-\frac{1}{3}$	-17

In the final stratum, [high] features associated to [-ATR] vowels are deleted, modeling the lowering of /ɪ, ʊ/ to [ɛ, ɔ]. This also derives the neutralization of

[a] and [ə]. Again, in this stratum, AGREE(ATR) must have a low weight to prevent its spreading regressively.

(22) Stratum 3

high ATR ↙ ↘ kɪbəfē	*[+high, -ATR] 2	Max(link)/high 1	$\mathcal{H}$
high ATR ↙ ↘ a. kɪbəfē			-2
ATR ↙ ↘ b. kɛbəfē		-1	-1

While this only sketches the analysis, it illustrates the intuition that non-local effects can be derived locally as a combination of serially ordered spreading processes.

4. CONCLUSION

Unbounded spreading poses a handful of analytic problems for Optimality Theory and Harmonic Grammar. This paper demonstrates that modeling autosegmental spreading as the redistribution of gradient activity (Tebay 2023) provides an empirically adequate approach that avoids these problems. Unlike in OT, spreading up to a blocker can be motivated with an AGREE constraint. Unlike in HG, cumulative faithfulness violations are not a problem as they remain constant. The model works in parallel frameworks as well as serial and non-local effects can be derived as spreading from the opposite direction. While these results are preliminary, we are optimistic that they can be built on and leave open the question of the model's empirical adequacy and typological predictions to future work.

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